What Challenges Do the Kidneys Face, and How Do They Overcome Them?

The kidneys are two bean-shaped organs about the size of a fist which reside just below the rib cage on either side of the spine. They constantly face problems with waste products from the blood, imbalanced electrolyte levels, and the task of regulating blood pressure. In this post, we will uncover how the kidneys deal with these issues on a general and microscopic level. [1]

Introduction:

The kidneys are commonly referred to as the 'master chemists' of the human body. This is because they filter, on average, approximately 190 litres of blood [1]. Not only do they filter a very high quantity of blood, but they do it at high efficiency. It takes 30 minutes for all the blood in the average human being to be completely filtered [2]!

How do the kidneys perform such amazing tasks at high volume and rate, while retaining the essential substances to a tee?

This post will go through how the kidneys work, the challenges which the kidneys face day-to-day and the physiological techniques that are used to overcome them.

How do the kidneys actually work?

The kidneys filter blood by first filtering all that is possible, and then returning the needed compounds back into the blood [3]. First, let's understand the path of which the blood takes throughout the kidneys ***(Figure 1)***:

Oxygenated blood flows into your kidneys through the renal artery via the aorta, where it then branches off into smaller and smaller branches eventually reaching the nephrons [3]. These nephrons are the micro-units which are responsible for filtering blood in the kidney. Once filtered, urine will be passed from the nephrons/tubules and into the ureter toward the bladder. Remaining filtered blood will then be passed out through the renal vein via the vena cava.

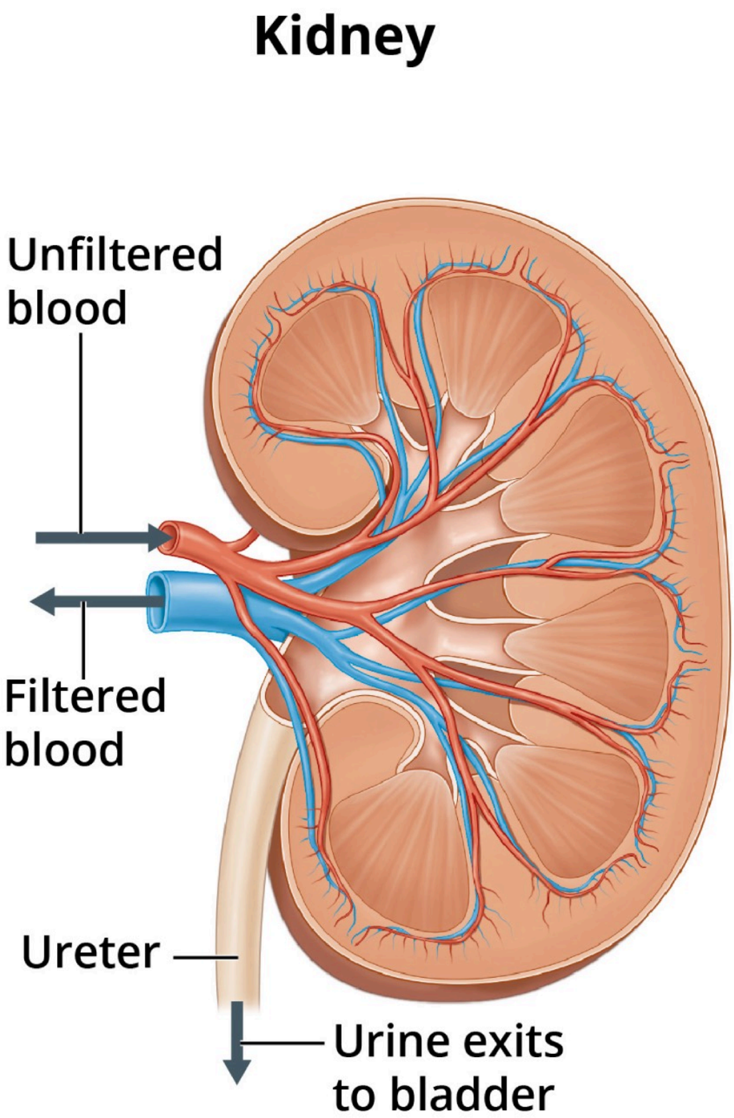

Figure 1*: Pathway of unfiltered blood through the kidneys. Source:* https://www.niddk.nih.gov/health-information/kidney-disease/kidneys-how-they-work

The filtration process is a relatively simple yet very microscopic process that takes place in ~ 900,000 to 1 million of individual nephrons throughout each kidney (nephron count can vary from 250,000 to 2.5 million per kidney yet 900,000 to 1 million is the general average) [4].

The nephrons are composed of a filter and a tubule (***Figure 2***); The filter is a cluster of tiny blood vessels known as the glomerulus, which separates small molecules which can pass through (like water or mineral ions), from the larger molecules (like glucose or urea) [3]. The tubule is responsible for allowing for the reabsorption of molecules which are in a deficit in the blood [3]. This is done as tubules run parallel beside a renal vein blood vessel, which allows for diffusion/osmosis of any molecules that are off higher concentration in the vein/blood than in the tubules [3].

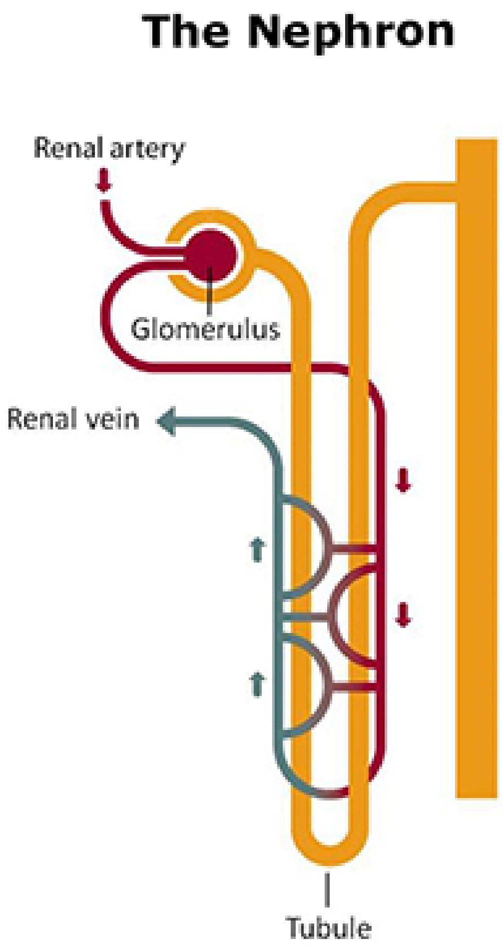

Figure 2*: Shows a simplified diagram of the nephron.*
https://www.niddk.nih.gov/health-information/kidney-disease/kidneys-how-they-work

Challenge 1: Conserving water while preventing waste build up

The first challenge a kidney must face is preventing unnecessary water loss from the blood while ensuring sufficient loss of waste. The way the kidneys achieve this is by manipulating the way water travels through osmosis and water potentials [5]. To understand how said process is completed, the anatomy of the nephron needs to be expanded:

As seen in Figure 3, the renal vein is looped around the beginning of the nephron, and is joined, by the capillaries, to the renal artery on the other side of the nephron past the first loop. This first loop is known as the loop of Henle. It is a ‘hair pin’ loop which consists of an ascending and descending limb [5]. Water which is absorbed through the glomerulus/glomerular capillary is passed down the proximal convoluted tubule (Figure 3) [5]. The proximal convoluted tubule is specially adapted to reabsorb ~85% of glomerular filtrate (absorbed product from the glomerular capillaries passed through the tubules), and hence, majority of water absorbed in the glomerulus is reabsorbed into the renal vein [5]. Ultrafiltration is the process of reabsorbing smaller molecules through a semi-permeable membrane [6], such as the proximal convoluted tubule and the loop of Henle [5].

The loop of Henle have a certain mechanism regarding the limbs and water potential, which allows them to extract water from the urine and globular filtrate, however, i believe the level of detail for a fully fledged explanation is unnecessary for the scale of this post, hence will be heavily simplified [5]; the **impermeable** ascending limb of the loop of Henle pumps out Na^+ ions, establishing a lower water potential in the interstitial region/fluid of the kidney (between descending and ascending limbs) [5]; the water from the globular filtrate and the urine in the collecting duct are then pulled out via osmosis and moved into the renal vein [5].

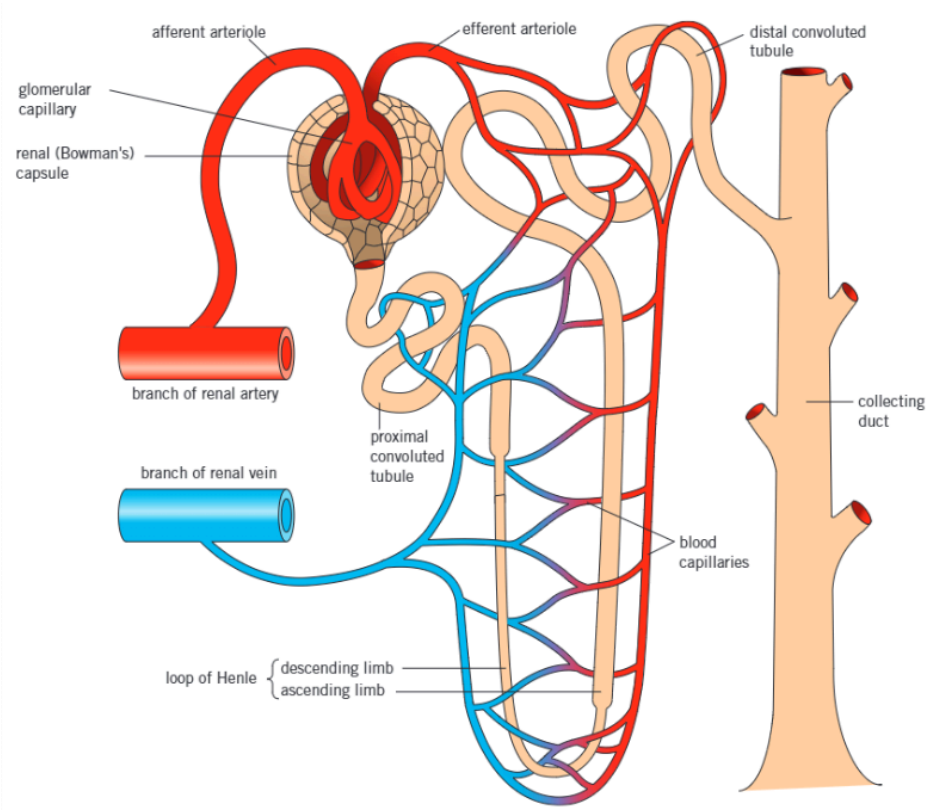

Figure 3: Kerboodle: A-level Sciences for AQA science; AQA Biology Year 2; https://www.kerboodle.com/app

Finally, the involvement of the antidiuretic hormone (ADH). ADH is a hormone secreted by the hypothalamus and stored in the pituitary gland [7]. While the collecting duct (Figure 3) is in the presence of ADH, it becomes permeable to water and solutes, meaning the ions present in the interstitial fluid move into the urine/ducts where urine becomes concentrated [7]. Inversely, in absence of ADH, ducts become impermeable to solutes and water, resulting in dilute urine [7]. When dehydrated, the interstitial fluid retains a higher potential than that of the blood [7]. Hence, when ADH is present and the ducts become permeable to water, water will move to the interstitial space and into the bloodstream via osmosis (and vice versa) [7].

As you can see, there are many different mechanisms that take place in the millions of neurons in a single kidney, determining water concentration of the blood and urine every single second. Isn't it fascinating, how such microscopic structures can perform such refined procedures in such a short fraction of time at such high efficiency and quality every day for years without fail?

Challenge 2: Maintaining an electrolyte balance

There are many vital electrolytes in the body which ensure the basic operations of life, such as maintenance of electrical neutrality and involvement of generating action potentials [8]. As such, the kidneys must ensure that there is a balance of said electrolytes, leading to different processes for a plethora of electrolytes [8]:

Sodium (Na/Na^+ in blood [9]): Na is responsible for maintaining extracellular fluid volume and involved in regulating/affecting membrane potential of cells (conducting nerve impulses) [8]. The proximal tubule (figure 3) is responsible for the majority of sodium reabsorption, whereas the rest takes place in the convoluted ducts [8]. Any remaining Na in filtrate may be reabsorbed at the distal convoluted tubule [8].The amount reabsorbed depends on certain hormones such as aldosterone [8].

Potassium (K/K^+ in blood [9]): Potassium is used mainly as an intracellular ion, in structures such as the sodium-potassium which regulates the homeostasis between Na^+ & K^+ ions [8]. Reabsorption of potassium occurs at the proximal convoluted tubule and the ascending limb of the loop of Henle [8,5](Figure 3). Potassium can also be secreted back into the urine via the distal convoluted tubule, where the greater the presence of aldosterone the more the secretion [8].

Magnesium (Mg/Mg^{2+} in blood [9]): Mg is an intracellular cation which is mainly involved in adenosine triphosphate (ATP) metabolism, the proper functioning of muscles and neurotransmitter release [8]. Assuming a normal Glomerular Filtration Rate (GFR), the kidney filters approximately 2000–2400 mg of magnesium per day [10]. The majority of magnesium is reabsorbed in the ascending limb of the loop of Henle (40% - 70%), where the rest is absorbed in the proximal convoluted tubule [10].

Final Challenge: Regulating Blood Pressure [11, 12]

Shocking to many, the kidney's purpose in the blood is not only balancing and filtering but it also pertains to blood pressure! This is especially important as the human blood’s pH can only safely be around 7.35 to 7.45 [13]. Even slightly stepping outside of that region can be almost, or entirely, fatal [13]! The system managing this is the Renin-Angiotensin-Aldosterone System (RAAS). RAAS is a system of proteins, enzymes, hormones and reactions which regulate the volume of blood and its pH for the long term, which is different from the usual, constant measuring and balancing of electrolytes or filtrates usually linked to the kidneys. By increasing Na^+/Salt & water reabsorption in the kidneys as well as vascular tone (narrowness of blood vessels), the RAAS can control the pH of blood to its optimal value.

- Renin is the enzyme made by your kidneys to help control blood pH as well as Na^+/K^+ levels.

- Angiotensin (II) is a hormone which helps increase blood volume, blood pressure and Na^+ levels.

- Aldosterone is a hormone made by the adrenal glands; It also helps maintain Na^+/K^+.

Steps for managing blood pressure [11]:

1. When your blood pressure falls, your kidneys release renin into your bloodstream.
2. Renin splits angiotensinogen into pieces. Angiotensinogen is a protein your liver makes and releases. One piece is the hormone angiotensin I.
3. Angiotensin I is inactive (doesn’t cause any effects). It flows through your bloodstream. Angiotensin-converting enzyme (ACE) in your lungs and kidneys split the angiotensin I into pieces. One of those pieces is Angiotensin II. Angiotensin II is an active hormone.
4. Angiotensin II causes the muscular walls of small arteries to narrow. This increases blood pressure. Angiotensin II triggers your adrenal glands to release aldosterone. It also triggers your pituitary gland to release antidiuretic hormone (ADH or vasopressin).
5. Together, aldosterone and ADH cause your kidneys to hold onto (retain) sodium. Aldosterone also causes your kidneys to release potassium through your pee.
6. The increase in sodium in your bloodstream causes your body to hold onto water. More water increases blood volume and blood pressure. This completes the renin-angiotensin-aldosterone system.

Conclusion:

Such complex and coordinated processes take place in your kidneys every second managing so many different quantities and values all to keep your body functioning and yet the kidneys are mainly known for its use to produce urine. The kidneys' role is million times larger than what you see here in this document, and the scale of processes likely exceeds what one may call microscopic, which is my reason for appreciating the kidneys to such an extent. The very fact that such precision and constant management for filtrates/pH/volume etc is even achievable and in ~1 million nephrons every minute of every day was unfathomable to me after first learning about it in year 10. Now that we have explored a more complex understanding of the kidneys capabilities, I hope I have managed to inspire you into looking further into other organs of the human body, or for that matter, the kidneys as well.

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